SOFTWARE REQUIREMENTS SPECIFICATION

Starlink Wi-Fi Subscription Portal

Captive-portal access control with Fapshi mobile money payments on MikroTik

Document	Software Requirements Specification (SRS)
Version	1.2 (merged baseline)
Status	Refined draft for development review
Date	1 October 2026
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Structure follows IEEE 830 and ISO/IEC/IEEE 29148 conventions.

Document Control

Revision history

Version	Date	Author	Description
1.0	1 October 2026	Mbuh Favour	First complete draft of the SRS.
1.1	1 October 2026	Mbuh Favour	Fixed the business rules: two plans, prices, clock start, local hosting.
1.2	1 October 2026	Mbuh Favour	Merged baseline. Keeps the full structure and detail of v1.0 and adopts the fixed decisions, payment-integrity rules, per-failure outage handling and extra acceptance criteria of v1.1.

What changed in version 1.2

- Plans are now fixed: 10-Day Plan at 2,000 XAF and 30-Day Plan at 5,000 XAF. No other plans are sold in the first release.
- The subscription clock starts at confirmed payment (closes OD-1 of v1.0).
- The portal server runs locally on the LAN, with no cloud hosting, domain or public IP required (closes OD-3 of v1.0).
- The server, not the browser, decides the price. A payment whose amount does not match the plan is rejected.
- Outage handling is split into Starlink, Fapshi, portal-server and MikroTik failures.
- Database business rules, a devices table, an order-applied marker and sample plan data were added to the data model.
- Acceptance criteria grew from 12 to 19 and every criterion is mapped to requirements.
- The traceability appendix was corrected so that it points only to requirement groups that exist.
- The plan-management admin page was removed, because plans are fixed for this release.

Document conventions

Requirements are written with the word shall for mandatory behaviour and should for recommended behaviour. Each requirement has a unique identifier such as FR-PY-03 (functional requirement, payment group, item 3). Priorities follow the MoSCoW scheme.

Priority	Meaning
Must	Essential for the first release. The system is not acceptable without it.
Should	Important and expected, but the system can launch without it if time is short.
Could	Desirable improvement that is included only if time and budget allow.
TBD	Value or decision not yet fixed. Listed in section 2.7 (Open decisions).

Table of Contents

Revision history . . . 2
What changed in version 1.2 . . . 2
Document conventions . . . 2
1. Introduction . . . 6
1.1 Purpose . . . 6
1.2 Scope . . . 6
1.3 Intended audience . . . 7
1.4 Definitions, acronyms and abbreviations . . . 8
1.5 References . . . 8
1.6 Document overview . . . 9
2. Overall Description . . . 9
2.1 Product perspective . . . 9
2.2 Product functions . . . 9
2.3 Core business model and subscription clock . . . 10
2.4 User classes and characteristics . . . 10
2.5 Operating environment . . . 11
2.6 Design constraints, cost and hosting . . . 11
2.7 Assumptions, dependencies and decisions . . . 11
3. System Architecture . . . 13
3.1 Component overview . . . 13
3.2 Subscription flow . . . 13
3.3 Administrator flows . . . 13
3.4 Separation of responsibilities . . . 14
3.5 Subscription lifecycle . . . 14
4. Functional Requirements . . . 15
4.1 Captive portal and login . . . 15
4.2 Plans . . . 15
4.3 Payment . . . 16
4.4 Access provisioning . . . 17
4.5 Device binding . . . 17
4.6 Anti-tethering and fair use . . . 18
4.7 Tenant status and help . . . 18
4.8 Administration portal . . . 19
4.9 Vouchers for cash payments . . . 19
4.10 Outage and failure handling . . . 19
4.11 Logging and reporting . . . 20

5. External Interface Requirements 20

5.1 User interfaces 20

5.2 Hardware interfaces 21

5.3 Software interfaces 21

5.4 Communications interfaces 22

6. Data Requirements 22

6.1 Main entities 22

6.2 Reference schema 23

6.3 Database business rules 24

6.4 Data protection and retention 24

7. Network and Router Configuration Requirements 25

8. Non-Functional Requirements 26

8.1 Performance and capacity 26

8.2 Security 26

8.3 Reliability and availability 26

8.4 Usability 27

8.5 Maintainability and portability 27

8.6 Legal and commercial 27

9. Use Cases 28

9.1 UC-01 Tenant buys a plan 28

9.2 UC-02 Tenant checks status 28

9.3 UC-03 Owner approves a tenant's laptop 28

9.4 UC-04 Owner re-binds a subscription 29

9.5 UC-05 Owner issues a cash voucher 29

9.6 UC-06 Starlink outage 29

9.7 UC-07 Subscription expiry 30

9.8 UC-08 Payment reconciliation 30

10. Technology Stack and Project Structure 30

11. Risk Analysis 32

12. Testing and Acceptance Criteria 32

12.1 Test approach 32

12.2 Acceptance criteria 34

13. Deployment and Operations 34

13.1 Deployment steps 34

13.2 Ongoing operations 35

14. Future Enhancements 35

15. Appendix A: Traceability Summary 36

16. Appendix B: Items to Confirm Before Development 36

17. Approval 36

1. Introduction

1.1 Purpose

This document specifies the requirements for the Starlink Wi-Fi Subscription Portal. The system lets the owner of a Starlink internet connection sell time-limited Wi-Fi access to tenants. A tenant who joins the Wi-Fi without having paid is shown a sign-in page where they can buy a plan using mobile money. After payment is confirmed, that tenant's device is given internet access for the purchased period, and nothing else needs to be done by hand.

The system lets a tenant:

1. Connect a phone, laptop or other supported device to the Wi-Fi network.
2. Be redirected to a captive portal if the device has no active subscription.
3. Choose one of the two subscription plans.
4. Pay with MTN Mobile Money or Orange Money through Fapshi.
5. Have the payment verified by the server.
6. Receive internet access automatically after confirmed payment.
7. Use the subscription only from the device it is bound to.
8. View the subscription status and expiry time.

The system also provides an administration portal through which the owner manages devices, subscriptions, payments, vouchers and exceptional cases.

The document is intended to be the single agreed description of what the system must do, so that design, development, network configuration, installation, testing and maintenance can all be checked against it.

1.2 Scope

In scope

- A captive portal on a MikroTik router that blocks internet access until a valid subscription exists.
- A tenant-facing website (plans, payment, status, help) built with plain HTML, CSS and JavaScript, mobile-first.
- A backend written in raw PHP (no framework) that talks to the Fapshi payment API and to the MikroTik router.
- MTN Mobile Money and Orange Money through Fapshi, subject to Fapshi API availability.
- Exactly two subscription plans: 2,000 XAF for 10 days and 5,000 XAF for 30 days.
- Automatic payment verification and automatic MikroTik provisioning after confirmed payment.
- Binding each subscription to a single device (MAC address) so one payment covers one device.
- Measures against tethering and hotspot sharing from a paid phone, and per-user speed and connection controls.
- An administration portal for the owner: device approval (for example tenant laptops), revoke, extend, re-bind, cash vouchers, payment records and audit log.

- Handling of Starlink outages, Fapshi failures, payment failures and router unavailability, with payment reconciliation.
- Local deployment of the portal server, database backups and recovery procedures.

Out of scope (first release)

- Data-volume quotas and usage-based billing.
- Multiple sites, buildings or routers managed from one portal.
- Native Android or iOS applications.
- Card payments, or payment providers other than Fapshi.
- Automatic recurring billing or auto-renewal.
- Self-service extra-device subscriptions (extra devices are approved by the owner).
- Referral systems and advanced revenue analytics.

1.3 Intended audience

Reader	Use of this document
Owner (project sponsor)	Confirms that the described behaviour, plans and controls match the business need.
Developer	Implements the portal, PHP backend and admin portal from the requirements, data model and interfaces.
Network technician	Configures the Starlink bypass, MikroTik hotspot, firewall rules and API access from sections 5 and 7.
Tester	Derives test cases from the functional requirements, use cases and acceptance criteria.

1.4 Definitions, acronyms and abbreviations

Term	Definition
Captive portal	A page that unauthenticated users are automatically redirected to before they can use the internet.
Hotspot (MikroTik)	The RouterOS feature that provides captive-portal login, user accounts, time limits and speed limits.
Walled garden	A list of addresses that unauthenticated devices are allowed to reach, for example the portal server.
MAC address	The hardware identifier of a device's network interface, used here to bind a subscription to one device.
Private (random) MAC	A per-network MAC address generated by modern phones for privacy. It is stable for a network unless the user resets it.
TTL	Time To Live, a counter in each IP packet that is reduced by one at every router hop. Tethered traffic shows a lower value.
Tethering	Sharing one device's internet connection with other devices using Wi-Fi hotspot, USB or Bluetooth.
Fapshi	Cameroonian payment service used to collect MTN Mobile Money and Orange Money payments.
MoMo / OM	MTN Mobile Money / Orange Money.
USSD	A mobile network service used to deliver the payment confirmation prompt to the tenant's phone.
CGNAT	Carrier-grade NAT. Starlink connections sit behind it, so the network cannot be reached directly from the internet.
Webhook	An HTTP request sent by Fapshi to the portal server when a payment changes state.
Voucher	A single-use access code generated by the owner for cash payments.
Subscription	A paid period of internet access bound to one device.
Reconciliation	Comparing Fapshi transactions with orders and subscriptions, and completing any that were missed.
XAF (FCFA)	Central African CFA franc, the currency of all prices in this system.
SRS	Software Requirements Specification.

1.5 References

- Fapshi developer documentation: payment initiation, transaction status and webhook behaviour (exact endpoints and fields to be confirmed against the current documentation before implementation).
- MikroTik RouterOS documentation: Hotspot, User Profiles, Walled Garden, Firewall and the RouterOS API (classic API and REST API in RouterOS 7).
- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018: guidance on requirements specifications.
- Starlink documentation on bypass mode and Ethernet adapters for the specific dish model in use.

1.6 Document overview

Section 2 describes the product as a whole. Section 3 describes the architecture and key flows. Section 4 lists the functional requirements. Section 5 covers external interfaces, section 6 the data model, section 7 the router configuration, and section 8 the non-functional requirements. Sections 9 to 13 contain use cases, the technology stack, risks, acceptance criteria and deployment notes. Section 14 lists possible future work, followed by the appendices and the approval table.

2. Overall Description

2.1 Product perspective

The system is a small, self-contained solution that sits between the Starlink internet source and the tenants. It has three cooperating parts: the MikroTik router (which enforces who is online), the portal server (which sells plans and talks to Fapshi) and the tenants' and owner's browsers. The Starlink router is placed in bypass mode where supported, so that the MikroTik becomes the only gateway for tenant devices.

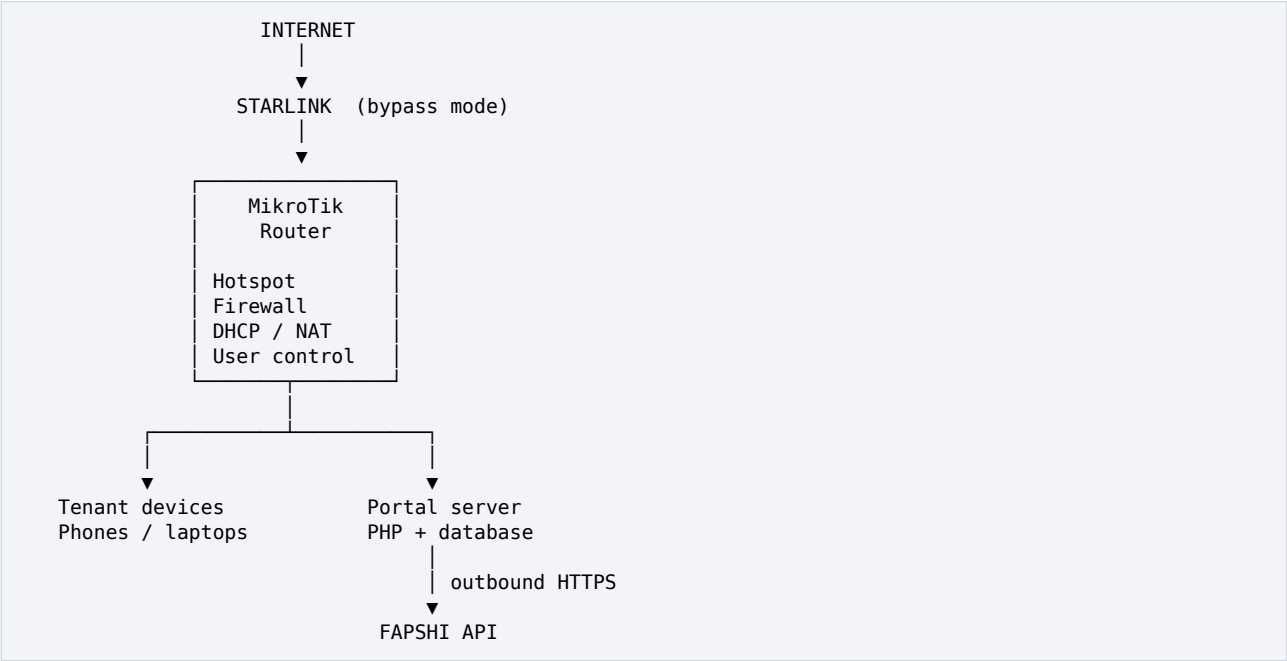


Figure 1. System context and main connections.

The MikroTik is the device responsible for enforcing internet access. The portal server handles business logic, payment processing, subscription management and communication with the MikroTik. The website is never the only mechanism that prevents internet access.

2.2 Product functions

- Intercept unpaid devices and show a sign-in page with a Buy a plan option.
- Offer the two fixed plans and show their prices and durations.
- Collect a mobile money payment through Fapshi and confirm it server-side, including the amount.
- Create a time-limited hotspot account bound to the paying device, and log the device in automatically.

- Allow only the paying device to use the subscription.
- Block tethering and apply per-user speed and connection limits.
- Let tenants see their plan, expiry and payment status.
- Give the owner an admin portal to approve extra devices (for example laptops), revoke or extend access, re-bind a changed device, issue cash vouchers and review payments.
- Warn tenants and disable purchases when the Starlink connection is down.

2.3 Core business model and subscription clock

The system provides exactly two plans. The prices and durations are fixed for this version.

Plan	Price	Validity
10-Day Plan	2,000 XAF	10 consecutive days
30-Day Plan	5,000 XAF	30 consecutive days

The subscription begins immediately after the server confirms the payment. For example, a 10-day subscription confirmed on 1 October at 14:30 becomes active at about 14:30 and expires 10 days later. The system stores the start time and expiry time explicitly. The days are calendar days and do not depend on how long the device is online.

The owner may change plans, prices or durations only through an approved change to this document.

2.4 User classes and characteristics

User class	Description	Technical level
Tenant	Resident who buys Wi-Fi access. Most tenants are students who use a single phone, on low-end to mid-range Android and iPhone models. Some also have a laptop.	Basic. Needs a very simple, mobile-first flow.
Owner / ad ministrator	Person who owns the Starlink connection and the router. Approves extra devices, handles cash payments and resolves problems.	Moderate. Comfortable with web forms, not necessarily with router settings.
Technician / developer	Installs and maintains the router configuration and the portal server.	Advanced.

2.5 Operating environment

Component	Environment
Internet source	Starlink dish with its router in bypass mode where supported. An Ethernet adapter may be required for some dish models.
Gateway	MikroTik router running RouterOS 7 (for example hAP ac2 or hAP ax2), with the Hotspot feature and API service enabled. Exact model to be confirmed (OD-4).
Portal server	Always-on small computer on the LAN (laptop, mini PC or Raspberry Pi) running Linux or Windows, a web server (Apache or Nginx) and PHP 8.x. Local hosting on the LAN is the fixed deployment for the first release.
Database	SQLite initially, or MySQL/MariaDB if required, on the portal server.
Tenant devices	Any current mobile or desktop browser that supports captive-portal detection.
Power	Mains power with a UPS or battery backup for the router and portal server.

2.6 Design constraints, cost and hosting

- The backend shall be written in raw PHP without a framework, because it is the language the developer is familiar with.
- The tenant frontend shall use plain HTML, CSS and JavaScript only, with about four pages.
- Payments shall use Fapshi. The Fapshi API credentials shall never be exposed to browsers.
- The router platform is MikroTik. Access control is performed by the MikroTik Hotspot, not by the website.
- The portal shall operate without recurring hosting costs when the MikroTik, Starlink connection and local server hardware already exist. A VPS, cloud hosting, a domain name, a public IP address and an external hosting provider shall not be required.
- The portal server communicates with Fapshi through outbound HTTPS only, so the system works behind Starlink CGNAT without incoming connections.
- Transaction fees charged by the payment provider are outside the software-development cost.

2.7 Assumptions, dependencies and decisions

Assumptions and dependencies

- The Starlink service is active and the dish has a clear view of the sky.
- Fapshi remains available and its API stays compatible with the integration described here.
- Tenants have an active MTN or Orange mobile money account and mobile network coverage to receive the payment prompt (USSD does not need mobile data).
- Tenants' devices keep a stable MAC address for the Wi-Fi network while a subscription is active.
- The portal server has a stable local IP address and can reach the internet through the MikroTik.
- The router clock is correct (time synchronisation is enabled), because expiry depends on it.

Decisions fixed in this version

Item	Decision
Number of plans	Exactly 2
10-Day Plan	2,000 XAF for 10 days
30-Day Plan	5,000 XAF for 30 days
Subscription start	Immediately after confirmed payment
Payment provider	Fapshi (MTN MoMo and Orange Money)
Gateway and internet	MikroTik and Starlink
Backend and frontend	Raw PHP; HTML, CSS and JavaScript
Hosting	Local server on the LAN. No cloud, domain or public IP required.
Extra devices	Added by the owner only. Tenant self-service is out of scope.
Not included	Card payment, recurring billing, multi-site support, data-volume billing
Device model	One authorised device per subscription

Open decisions

ID	Decision needed	Current default / proposal
OD-1	Speed profile for each plan.	TBD by the owner. Use the planning formula in section 8.1 after measuring Starlink throughput.
OD-2	How Fapshi payments are collected: hosted checkout page or direct charge to the phone number.	Preferred: direct charge, so tenants never leave the portal. Confirm with Fapshi documentation.
OD-3	How payment confirmation is received: webhook, polling, or both.	Proposed: polling as the default, with webhook if a tunnel is added.
OD-4	Router model and RouterOS version.	TBD. A RouterOS 7 device is assumed.
OD-5	Starlink dish and router generation, and the Ethernet arrangement.	TBD.
OD-6	Portal server hardware.	TBD. Must be selected and tested.
OD-7	Interface languages.	Proposed: English and French.
OD-8	Data retention period for payment and device records.	Proposed: 12 months, then anonymise or delete.

3. System Architecture

3.1 Component overview

Component	Responsibility
Starlink dish and router	Provides the internet connection. The router runs in bypass mode and does not control tenants.
MikroTik router	Wi-Fi and wired LAN, DHCP, NAT, Hotspot captive portal, per-user profiles, expiry, walled garden, TTL and connection limiting, and the API used by the portal server.
Tenant frontend	Static HTML, CSS and JavaScript pages served by the portal server: plans, payment, status and help.
PHP backend	Creates orders, calls Fapshi, confirms payments, provisions and removes hotspot users on the router, serves status data and runs scheduled reconciliation.
Admin portal	Password-protected PHP pages for the owner.
Database	Stores plans, orders, devices, subscriptions, vouchers, admins and audit entries.
Fapshi	Collects mobile money payments and reports their status.

3.2 Subscription flow

Step	Actor	Action
1	Tenant	Connects a device to the Wi-Fi. The MikroTik detects it has no active subscription.
2	MikroTik	Intercepts web traffic and shows the hotspot login page, with a Buy a plan button and a voucher field.
3	Tenant	Taps Buy a plan. The router redirects to the portal and passes the device MAC address and IP address.
4	Portal	Shows the two plans. The tenant selects one and enters a mobile money phone number.
5	Backend	Reads the authoritative price from the database, creates an order with a unique reference and status Pending, then asks Fapshi to start the payment.
6	Tenant	Approves the payment on the phone with the mobile money PIN.
7	Backend	Receives the Fapshi notification or polls Fapshi, then verifies the transaction status and amount directly with Fapshi.
8	Backend	If the payment is successful, the amount matches the plan and the transaction has not been processed before, creates the subscription and a hotspot user named after the device MAC address with the plan's limits.
9	MikroTik	Logs the device in automatically through MAC login. Internet access begins.
10	Portal	Status page shows the plan and expiry. The router enforces expiry and removes access at the end.

3.3 Administrator flows

- Approve a device: the tenant connects a laptop, the owner opens the admin portal, picks the device from the list of connected but not logged-in devices, selects tenant and plan, and approves it.
- Cash voucher: the owner generates a single-use code, hands it to the tenant, who enters it on the login page. The code binds to that tenant's device.
- Re-bind: when a tenant's device MAC changes, the owner moves the active subscription to the new MAC address and the old one loses access.
- Revoke or extend: the owner stops a subscription immediately or adds time to it.

3.4 Separation of responsibilities

Website and backend	MikroTik
Display plans Create orders Communicate with Fapshi Verify payments Create subscriptions Store records Communicate with the MikroTik Provide tenant status and administration Record audit events	Provide Wi-Fi and LAN Redirect unauthenticated devices Enforce access and expiry Apply user profiles Apply speed and connection limits Restrict unauthorised devices Apply firewall and anti-tethering rules

3.5 Subscription lifecycle

State	Meaning	Possible next states
Pending	Order created, waiting for Fapshi confirmation.	Paid, Failed, Expired order
Paid	Payment confirmed and amount verified, router user not yet created (for example router unreachable).	Active, Paid (retry)
Active	Hotspot user exists and the paid period is running.	Expired, Revoked, Active (extended or re-bound)
Expired	Purchased period ended. Router user removed or disabled.	None (tenant buys again)
Revoked	Stopped by the owner.	None (tenant buys again or owner re-approves)
Failed	Payment declined, cancelled or timed out.	None (tenant may retry with a new order)

4. Functional Requirements

4.1 Captive portal and login

ID	Requirement	Priority
FR-CP-01	The router shall intercept every unauthenticated device and redirect its web traffic to the hotspot login page.	Must
FR-CP-02	The system shall prevent an unpaid device from accessing the external internet.	Must
FR-CP-03	The login page shall present a Buy a plan button and a field to enter an existing voucher code.	Must
FR-CP-04	The redirect to the portal shall carry the device MAC address and IP address as parameters.	Must
FR-CP-05	The walled garden shall allow unauthenticated devices to reach the portal server and nothing else, unless hosted checkout is used, in which case the required Fapshi domains shall also be allowed.	Must
FR-CP-06	The login page and portal pages shall be light enough to load quickly on low-end phones.	Must
FR-CP-07	The login page shall work with the automatic captive-portal pop-up on Android, iOS, Windows and macOS, and shall also be reachable manually by opening a plain HTTP address.	Should
FR-CP-08	The login page shall show a service-down notice when the Starlink connection is unavailable (see FR-OU-02).	Should

4.2 Plans

ID	Requirement	Priority
FR-PL-01	The system shall provide exactly two plans and shall not display or sell any other plan in the first release.	Must
FR-PL-02	The 10-Day Plan shall cost 2,000 XAF and provide 10 days of access.	Must
FR-PL-03	The 30-Day Plan shall cost 5,000 XAF and provide 30 days of access.	Must
FR-PL-04	The plans page shall list both plans with price and duration, using clear language.	Must
FR-PL-05	The duration of every plan shall be counted from the moment payment is confirmed (see section 2.3).	Must
FR-PL-06	The browser shall not determine or modify the plan price. The backend shall read the authoritative price from the database.	Must
FR-PL-07	The system shall reject a payment whose amount does not match the selected plan.	Must
FR-PL-08	Changing a plan in the future shall not alter subscriptions already purchased.	Must
FR-PL-09	The two plans shall be created in the database at deployment. The first release has no plan-management screen.	Should

Plan	Duration	Price (XAF)	Speed profile	Devices
10-Day Plan	10 days	2,000	TBD (OD-1)	1
30-Day Plan	30 days	5,000	TBD (OD-1)	1

Table 4.2. The fixed plans. Speed profiles are decided by the owner (OD-1).

4.3 Payment

ID	Requirement	Priority
FR-PY-01	The tenant shall enter a mobile money phone number, which the system shall validate against the Cameroonian format before sending it to Fapshi.	Must
FR-PY-02	Before contacting Fapshi, the backend shall create an order record with a unique reference, the plan, the device MAC address, the amount and the status Pending.	Must
FR-PY-03	The backend shall start the payment with Fapshi, sending the correct amount in XAF, using credentials stored on the server only. Fapshi credentials shall never be exposed to tenant browsers.	Must
FR-PY-04	The backend shall confirm every payment by querying Fapshi for the transaction status. A webhook or a browser redirect alone shall never grant access.	Must
FR-PY-05	The system shall support confirmation by polling Fapshi, and by webhook where a public URL is available. Polling shall not overload Fapshi or the local server.	Must
FR-PY-06	Each Fapshi transaction identifier shall be stored as unique, so repeated notifications can never create two subscriptions.	Must
FR-PY-07	The portal shall show the tenant a clear pending screen while waiting, and a clear success or failure message afterwards.	Must
FR-PY-08	Orders that remain Pending beyond a configurable timeout (default 10 minutes) shall be marked Expired order, after a final status check with Fapshi.	Must
FR-PY-09	A tenant whose payment failed or was cancelled shall be able to retry without contacting the owner.	Should
FR-PY-10	The system shall never ask for, receive or store mobile money PINs.	Must
FR-PY-11	The server shall verify that the amount confirmed by Fapshi equals the price of the selected plan. On a mismatch no access shall be granted and the order shall be flagged for the owner.	Must
FR-PY-12	Failed, declined, cancelled and expired payments shall not activate internet access.	Must
FR-PY-13	Payment processing errors shall be recorded for troubleshooting.	Must

4.4 Access provisioning

ID	Requirement	Priority
FR-AP-01	A subscription shall only be created after confirmed payment, or through a valid voucher or an owner approval.	Must
FR-AP-02	Each subscription shall have a unique identifier and shall record its plan, device, start time and expiry time.	Must
FR-AP-03	After a confirmed payment, the backend shall create a hotspot user on the MikroTik named after the paying device's MAC address, with the plan's profile and expiry.	Must
FR-AP-04	The device shall be logged in automatically through MAC login, or shown a single-tap login button if automatic login is not possible.	Must
FR-AP-05	The router shall enforce expiry by itself, so access ends on time even if the portal server is off. Expiry shall be based on calendar time from the start time.	Must
FR-AP-06	Provisioning shall complete within 10 seconds of payment confirmation under normal conditions.	Should
FR-AP-07	If the router API is unreachable, the order shall be marked Paid, retried automatically at short intervals, and flagged on the admin dashboard.	Must
FR-AP-08	Expired hotspot users shall be removed or disabled by the router, and the portal database shall be updated by a scheduled job.	Should
FR-AP-09	A tenant who buys a second plan while one is active shall have the new duration added to the end of the existing subscription instead of getting a second one.	Should
FR-AP-10	The system shall not create more than one subscription, or apply more than one extension, from the same successful transaction.	Must

4.5 Device binding

ID	Requirement	Priority
FR-DB-01	One paid subscription shall authorise exactly one device, identified by its MAC address.	Must
FR-DB-02	A second device attempting to use the same subscription shall be shown the login page and shall have to pay or be approved separately.	Must
FR-DB-03	Each hotspot user shall be limited to one simultaneous session.	Must
FR-DB-04	The system shall store the device MAC address with each order and subscription, and show the associated device to the owner.	Must
FR-DB-05	The owner shall be able to re-bind an active subscription to a new MAC address. The previous device shall lose authorisation.	Must
FR-DB-06	Every re-bind shall be recorded in the audit log.	Must
FR-DB-07	If a tenant's MAC address changes (for example after resetting the Wi-Fi network on the phone), the portal shall show a clear message to contact the owner.	Should
FR-DB-08	The system shall use the network-specific MAC address presented by the device when the subscription is activated, and shall not assume that the physical hardware MAC is available.	Must
FR-DB-09	The help page shall explain that resetting the private Wi-Fi address can make a device appear as a different device.	Must

4.6 Anti-tethering and fair use

ID	Requirement	Priority
FR-AT-01	The system shall apply measures to restrict common Wi-Fi, USB and Bluetooth tethering.	Must
FR-AT-02	The router shall apply a TTL filter on the hotspot network that accepts the normal TTL values 64 and 128 and drops packets with other values, which indicate traffic forwarded through another device.	Should
FR-AT-03	The filter shall also cover IPv6 traffic, or IPv6 shall be disabled on the hotspot network.	Should
FR-AT-04	Every plan profile shall include a per-user speed limit.	Must
FR-AT-05	Every plan profile shall include a per-user connection limit.	Should
FR-AT-06	Directly connected laptops approved by the owner shall not be affected by the filter.	Must
FR-AT-07	The owner shall be able to switch the filter on or off on the router without changing the portal software.	Should
FR-AT-08	Anti-tethering shall be judged by observed behaviour, not by one mechanism alone, and shall be tested on supported Android and iOS devices.	Must
FR-AT-09	The system shall be documented as providing best-effort control only: a VPN or a modified TTL value can bypass the filter.	Must

4.7 Tenant status and help

ID	Requirement	Priority
FR-ST-01	A logged-in tenant shall be able to see the current plan, payment status, start time, expiry date and time, and remaining time.	Must
FR-ST-02	The status page shall identify the tenant only through the device that is connected, not through a typed phone number. It shall not expose another tenant's information.	Must
FR-ST-03	The help page shall explain how to pay, what to do if payment succeeded but there is no access, the one-device rule, private MAC addresses, and how to contact the owner.	Must
FR-ST-04	Tenants shall be able to see their last few payments and their reference numbers.	Could
FR-ST-05	Pages shall be available in the languages decided in OD-7 (proposed English and French).	Should

4.8 Administration portal

ID	Requirement	Priority
FR-AD-01	Admin pages shall require login with a username and password. Passwords shall be stored as salted hashes.	Must
FR-AD-02	The portal shall list devices that are connected to the router but not logged in, showing MAC address, IP address and host name.	Must
FR-AD-03	The owner shall be able to approve a device from that list by choosing a tenant name and a plan. The system shall then create a MAC-bound hotspot user.	Must
FR-AD-04	The owner shall be able to revoke a subscription immediately, which also disconnects the active session.	Must
FR-AD-05	The owner shall be able to extend a subscription by a chosen amount of time.	Should
FR-AD-06	The owner shall be able to re-bind a subscription to another MAC address.	Must
FR-AD-07	The dashboard shall show active subscriptions, expired subscriptions, today's payments and revenue, pending or stuck payments, provisioning errors, and router and Fapshi status.	Should
FR-AD-08	The owner shall be able to search payments by phone number, reference, MAC address and date, and subscriptions by tenant or device.	Should
FR-AD-09	The owner shall be able to view authorised devices and revoke a device.	Must
FR-AD-10	Every administrative action shall be recorded in the audit log with time, administrator and details.	Must
FR-AD-11	The admin portal shall not be reachable by unauthenticated tenants through the walled garden, and shall accept connections only from the local network or configured addresses.	Must

4.9 Vouchers for cash payments

ID	Requirement	Priority
FR-VC-01	The owner shall be able to generate one or more vouchers, each for one of the two plans, and see whether a voucher has been used.	Must
FR-VC-02	A voucher shall be single-use. When redeemed on the login page, it shall bind the subscription to the redeeming device and become permanently unavailable.	Must
FR-VC-03	Vouchers shall have an expiry date after which they cannot be redeemed.	Could
FR-VC-04	Vouchers shall be generated with enough randomness to prevent guessing, and failed redemption attempts shall be rate limited.	Should
FR-VC-05	Voucher generation and redemption shall be recorded in the audit log.	Must

4.10 Outage and failure handling

Each kind of failure is handled independently.

ID	Requirement	Priority
FR-OU-01	The portal server shall regularly check that the internet connection and the Fapshi API are reachable.	Should
FR-OU-02	Starlink outage: the portal shall not present internet access as available, shall show a service-temporarily-down notice and shall disable the Pay button.	Must

ID	Requirement	Priority
FR-OU-03	If a payment was started just before an outage, the system shall reconcile it with Fapshi when the connection returns and provision access if the payment succeeded and the amount is correct.	Must
FR-OU-04	Fapshi outage: new payments shall never be marked successful without confirmation, the tenant shall see an appropriate message, and pending transactions shall be reconciled when the service returns.	Must
FR-OU-05	Portal server outage: the router shall continue to enforce existing subscriptions until their expiry. The portal server is not needed for every packet after provisioning.	Must
FR-OU-06	MikroTik outage: no tenant access is possible through the system. The application shall record provisioning failures when it can detect them.	Must
FR-OU-07	Cash vouchers and owner approvals shall keep working while the internet is down, as long as the portal server and router are running.	Should

4.11 Logging and reporting

ID	Requirement	Priority
FR-LG-01	The system shall log payment events and status changes, Fapshi responses, subscription creation, router API calls and errors, including MikroTik provisioning errors.	Must
FR-LG-02	Secrets and API keys shall never be written to logs.	Must
FR-LG-03	The system shall provide a reconciliation report that compares Fapshi transactions with subscriptions created.	Should
FR-LG-04	The owner shall be able to export payments and subscriptions as CSV for a chosen date range.	Could

5. External Interface Requirements

5.1 User interfaces

Tenant pages (plain HTML, CSS and JavaScript, mobile-first):

Page	Purpose	Key elements
Plans (index)	Show the two plans and start a purchase.	Plan cards with price and duration. Service-status banner. Purchase buttons. Language switch.
Checkout	Collect the phone number and show payment progress.	Selected plan and price, phone number field, payment instructions, pending animation, success and failure messages with Retry.
Status	Show the current subscription.	Plan name, payment status, start time, expiry, remaining time, device note.
Help	Support and rules.	Step-by-step payment guide, one-device rule, private MAC note, no-access troubleshooting, owner contact details.

Admin pages (PHP, for the owner):

Page	Purpose
Login	Authenticate the owner.
Dashboard	Active subscriptions, payments today, stuck payments, router and Fapshi status.
Devices	Connected and not-yet-logged-in devices with an Approve action, plus authorised devices.
Subscriptions	Search, revoke, extend, re-bind.
Payments	All orders and transactions with filters and CSV export.
Vouchers	Generate and list vouchers, with used or unused status.
Audit log	History of administrative actions.

Interface principles

- Large touch targets, readable font sizes and high contrast.
- Usable on low-end Android devices, iPhones and laptops.
- No dependence on large images, external fonts or unnecessary third-party scripts, because unpaid devices can reach only the portal server.
- Plain language messages that tell the tenant exactly what to do next.

5.2 Hardware interfaces

Item	Interface
Starlink dish and router	Ethernet from the Starlink router (in bypass mode) to the MikroTik WAN port. An Ethernet adapter may be needed for some dish models.
MikroTik router	WAN port to Starlink, bridged LAN ports and Wi-Fi to tenants, LAN port to the portal server.
Portal server	Ethernet to the MikroTik LAN. Static or reserved IP address.
UPS or battery backup	Powers the router and portal server during power cuts so subscriptions keep running.

5.3 Software interfaces

5.3.1 Fapshi payment API

Aspect	Description
Direction	Backend to Fapshi (outbound HTTPS). Optionally Fapshi to backend for webhooks.
Operations used	Initiate a payment, query transaction status, receive payment-status notifications.
Data sent	Amount in XAF, phone number, order reference, optional description.
Data received	Transaction identifier, status (for example created, successful, failed, expired), amounts and timestamps.
Authentication	API user and API key held in a server-side configuration file outside the web root.
Notes	Exact endpoints, field names, authentication, webhook behaviour, supported networks and status values must be confirmed against the current Fapshi documentation before implementation. The integration shall be isolated in one PHP module so that changes affect only that module.

5.3.2 MikroTik RouterOS API

Operation	Purpose
Add hotspot user	Create the MAC-named user with profile, limits and comment (order reference).
Remove or disable hotspot user	Revoke or expire a subscription.
Update user	Change expiry (extend) or MAC address (re-bind).
List hosts	Show connected devices that are not yet authorised (for the Devices page).
List active sessions	Show who is online and remove a session on revoke.
Read user and profile data	Reconcile router state with the database.
Check access	Connectivity test used for health monitoring.

The API shall be accessed by a dedicated low-privilege router account, using the encrypted API service or REST over HTTPS where possible, and restricted to the portal server's IP address. Router API credentials shall be stored outside the public web root and never exposed to tenants.

5.4 Communications interfaces

- Tenant devices to portal server: HTTP or HTTPS over the LAN, allowed through the walled garden.
- Portal server to MikroTik: RouterOS API over the LAN, restricted by firewall to the portal server address.
- Portal server to Fapshi: outbound HTTPS through the Starlink connection.
- Optional: a free tunnel service to give the portal a public HTTPS address if Fapshi webhooks are used. Polling works without it.

6. Data Requirements

6.1 Main entities

Entity	Description	Key relationships
plans	The two sellable plans with duration, price and router profile.	One plan has many orders and subscriptions.
orders	A payment attempt with Fapshi reference, phone number, device MAC, amount and status.	Belongs to a plan. Applied to at most one subscription.
devices	Devices seen on the network, with MAC address, description and first and last seen times.	Referenced by MAC address from orders, subscriptions and vouchers.
subscriptions	A granted access period bound to one MAC address, with explicit start and expiry times.	Belongs to a plan and optionally an order or voucher.
vouchers	Cash-payment codes with status.	Belongs to a plan. May be redeemed into a subscription.
admins	Owner accounts for the admin portal.	Referenced by the audit log.
audit_log	Record of administrative and system actions.	References an admin where applicable.

6.2 Reference schema

The schema below is a starting point written in generic SQL. Column types may be adjusted for SQLite or MySQL.

```
CREATE TABLE plans (
 id INTEGER PRIMARY KEY,
 name VARCHAR(60) NOT NULL,
 duration_days INTEGER NOT NULL,
 price_xaf INTEGER NOT NULL,
 router_profile VARCHAR(60) NOT NULL,
 max_devices INTEGER NOT NULL DEFAULT 1,
 is_active INTEGER NOT NULL DEFAULT 1,
 sort_order INTEGER NOT NULL DEFAULT 0
);

INSERT INTO plans (name, duration_days, price_xaf, router_profile, sort_order)
VALUES ('10-Day Plan', 10, 2000, 'profile_10d', 1),
       ('30-Day Plan', 30, 5000, 'profile_30d', 2);
```

```
CREATE TABLE orders (
 id INTEGER PRIMARY KEY,
 reference VARCHAR(40) NOT NULL UNIQUE,
 plan_id INTEGER NOT NULL,
 phone VARCHAR(20) NOT NULL,
 device_mac VARCHAR(17) NOT NULL,
 amount_xaf INTEGER NOT NULL,
 fapshi_trans_id VARCHAR(64) UNIQUE,
 status VARCHAR(20) NOT NULL DEFAULT 'pending',
 subscription_id INTEGER,
 created_at DATETIME NOT NULL,
 updated_at DATETIME NOT NULL,
 confirmed_at DATETIME,
 FOREIGN KEY (plan_id) REFERENCES plans (id)
);

CREATE TABLE devices (
 id INTEGER PRIMARY KEY,
 mac_address VARCHAR(17) NOT NULL UNIQUE,
 description VARCHAR(80),
 first_seen DATETIME NOT NULL,
 last_seen DATETIME NOT NULL,
 status VARCHAR(20) NOT NULL DEFAULT 'unauthorised'
);
```

```
CREATE TABLE subscriptions (
 id INTEGER PRIMARY KEY,
 order_id INTEGER UNIQUE,
 voucher_id INTEGER,
 plan_id INTEGER NOT NULL,
 device_mac VARCHAR(17) NOT NULL,
 tenant_name VARCHAR(80),
 status VARCHAR(20) NOT NULL DEFAULT 'active',
 starts_at DATETIME NOT NULL,
 expires_at DATETIME NOT NULL,
 created_by VARCHAR(20) NOT NULL DEFAULT 'payment',
 created_at DATETIME NOT NULL,
 FOREIGN KEY (plan_id) REFERENCES plans (id)
);

CREATE TABLE vouchers (
 id INTEGER PRIMARY KEY,
 code VARCHAR(20) NOT NULL UNIQUE,
 plan_id INTEGER NOT NULL,
 status VARCHAR(20) NOT NULL DEFAULT 'unused',
 redeemed_by_mac VARCHAR(17),
 redeemed_at DATETIME,
 expires_at DATETIME,
 created_at DATETIME NOT NULL,
 FOREIGN KEY (plan_id) REFERENCES plans (id)
);
```

```
CREATE TABLE admins (
 id INTEGER PRIMARY KEY,
 username VARCHAR(40) NOT NULL UNIQUE,
 password_hash VARCHAR(255) NOT NULL,
 status VARCHAR(20) NOT NULL DEFAULT 'active',
 created_at DATETIME NOT NULL
);

CREATE TABLE audit_log (
 id INTEGER PRIMARY KEY,
 admin_id INTEGER,
 action VARCHAR(60) NOT NULL,
 target VARCHAR(80),
 details TEXT,
 created_at DATETIME NOT NULL
);
```

Listing 6.1. Reference database schema.

6.3 Database business rules

The database or the application shall enforce the following rules.

1. Each Fapshi payment reference shall be unique.
2. A successful payment shall be applied at most once: it shall not create more than one subscription or extension.
3. A subscription shall have exactly one plan and one associated device.
4. An active subscription shall have a valid start time and expiry time.
5. An expired subscription shall not remain active.
6. A voucher shall not be redeemable more than once.
7. The confirmed payment amount shall match the plan price.
8. A subscription shall not be activated without confirmed payment, a valid voucher or an owner approval.
9. Every administrative re-bind shall be logged.

6.4 Data protection and retention

- The system shall store only the data needed to run the service: phone number, device MAC address, plan, payment reference and timestamps.
- Mobile money PINs, card data and tenant passwords shall never be handled or stored.
- Admin passwords shall be stored using PHP's password hashing functions.
- The proposed retention period is 12 months, after which records are deleted or anonymised according to the owner's final policy (OD-8).
- The owner shall handle tenant data in line with applicable Cameroonian personal-data legislation (to be verified), and a short privacy notice shall be shown on the help page.
- Database backups shall be taken regularly and stored away from the portal server.

7. Network and Router Configuration Requirements

This section lists what the router must be configured to do. It is deliberately written as requirements and not as command scripts, because exact settings depend on the router model and RouterOS version.

ID	Area	Requirement
NR-01	Starlink	The Starlink router shall be in bypass mode where supported, so the MikroTik is the only gateway for tenants. Use the Ethernet adapter where the dish model requires it.
NR-02	WAN and NAT	The MikroTik WAN shall obtain its address from Starlink by DHCP, with NAT masquerade for the LAN.
NR-03	LAN	All tenant Wi-Fi and wired ports shall belong to one bridge with its own DHCP pool. The portal server shall have a reserved address.
NR-04	Hotspot	The Hotspot shall run on the tenant bridge, with MAC-address login enabled in addition to username and password.
NR-05	User profiles	One profile per plan, each with a rate limit, one shared user and a connection limit.
NR-06	Expiry	Subscription expiry shall be enforced by the router on calendar time from the start time (for example a scheduler that removes the hotspot user at its expiry timestamp), not by online time. The router clock shall be synchronised by NTP.
NR-07	Walled garden	Unauthenticated devices shall reach only the portal server and, if hosted checkout is used, the listed Fapshi domains.
NR-08	Login page	The hotspot login page shall be customised with Buy a plan and voucher entry, and shall forward MAC and IP to the portal.
NR-09	TTL filter	A firewall rule on forwarded hotspot traffic shall drop packets whose TTL is not 64 or 128. IPv6 shall be covered or disabled.
NR-10	API access	A dedicated API user with only the permissions needed shall exist. The API service shall accept connections only from the portal server's IP.
NR-11	Management security	Default accounts shall be changed or disabled. Unused services shall be disabled. Router management shall not be reachable from tenant addresses.
NR-12	Backups	A router configuration export shall be saved after every change and kept off the router.
NR-13	Updates	RouterOS shall be updated in planned maintenance windows after reading release notes.
NR-14	Persistence	Hotspot users and profiles shall survive a router restart, so existing subscriptions stay valid.

8. Non-Functional Requirements

8.1 Performance and capacity

ID	Requirement
NFR-PF-01	Portal pages shall load in under 3 seconds over the local Wi-Fi.
NFR-PF-02	Provisioning shall complete within 10 seconds of confirmed payment under normal conditions, subject to Fapshi and MikroTik response time.
NFR-PF-03	The portal shall support at least 60 simultaneous tenant sessions and 100 registered tenants (design target, to be adjusted by the owner).
NFR-PF-04	Expired subscriptions shall lose access within 2 minutes of their expiry time.
NFR-PF-05	Payment-status polling shall not unnecessarily overload Fapshi or the local server.

Speed limits per plan should be chosen from the available Starlink bandwidth. A simple planning formula for the guaranteed share per active user is:

$$B_{user} = (B_{total} \times u) / N_{active}$$

where B_{total} is the measured Starlink throughput, u is a utilisation factor (for example 0.8, leaving headroom for overhead and bursts), and N_{active} is the number of tenants online at the same time. Example only: B_{total} = 100 Mbps, u = 0.8 and N_{active} = 40 gives B_{user} = 2 Mbps.

Starlink throughput varies with weather and obstructions, so B_{total} should be measured at different times of day before plan speeds are fixed (OD-1).

8.2 Security

ID	Requirement
NFR-SC-01	Fapshi credentials and router passwords shall be stored in a configuration file outside the web root and never sent to the browser.
NFR-SC-02	All database access shall use prepared statements to prevent SQL injection.
NFR-SC-03	All output to pages shall be escaped to prevent cross-site scripting.
NFR-SC-04	Admin sessions shall use secure session settings, expire after inactivity, and be protected against cross-site request forgery and session hijacking.
NFR-SC-05	Login attempts to the admin portal and voucher redemption shall be rate limited.
NFR-SC-06	Webhook requests shall never be trusted on their own. Every payment shall be verified with Fapshi, including amount and transaction identity, before access is granted. Replayed notifications shall have no effect.
NFR-SC-07	The router API shall be reachable only from the portal server and shall use an account with minimal permissions.
NFR-SC-08	The admin portal shall be reachable only from the local network or configured addresses, and shall use HTTPS where possible.
NFR-SC-09	Error messages shown to tenants shall not reveal internal paths, queries or credentials.
NFR-SC-10	Input shall be validated, and parameter manipulation shall not allow unauthorised subscription activation.

8.3 Reliability and availability

- The router and portal server shall start automatically after a power cut, with no manual steps.
- Database records shall be preserved across a portal-server restart.
- Existing subscriptions shall keep working on the router even when the portal server is off.
- A UPS or battery backup should keep the MikroTik, portal server and related network equipment running through short power cuts.
- The database shall be backed up automatically and the backup restored in a test at least once.
- Provisioning failures shall be retried automatically and shown on the dashboard, never lost silently.
- Pending payments shall be reconciled after connectivity returns.

8.4 Usability

- A tenant shall be able to buy a plan in no more than four taps and one phone number entry, without technical knowledge.
- Instructions shall be written in simple language in the languages decided in OD-7.
- The interface shall work on small screens and low-end browsers without requiring installation.
- The admin portal shall let the owner approve a laptop in under one minute.

8.5 Maintainability and portability

- Code shall be organised into separate modules for database, Fapshi, MikroTik, authentication, subscriptions, orders and administration, and the Fapshi and router integrations shall each live in one module.
- A change in the Fapshi API shall preferably require changes only in the Fapshi module.
- Settings such as timeouts and addresses shall be in the database or a configuration file, not hard-coded in several places.
- Code shall be commented where behaviour is not obvious, with one statement per line for easy reading.
- The portal shall run on any Linux or Windows machine with PHP 8.x, a web server and SQLite or MySQL, so the server can be replaced easily.

8.6 Legal and commercial

- A short terms-of-use and acceptable-use notice shall be shown on the help page.
- A refund and dispute procedure shall be defined by the owner and described on the help page.
- Fapshi fees and any applicable taxes shall be considered by the owner when reviewing plan prices.

9. Use Cases

9.1 UC-01 Tenant buys a plan

Actor	Tenant
Preconditions	Device is connected to the Wi-Fi, has no active subscription, and the service is up.
Main flow	1. Login page appears. 2. Tenant taps Buy a plan. 3. Tenant selects the 10-Day or 30-Day Plan. 4. Tenant enters a mobile money number. 5. System creates an order at the server-side price and starts the payment with Fapshi. 6. Tenant approves on the phone. 7. System confirms the payment and amount with Fapshi. 8. System creates the subscription and the MAC-bound hotspot user. 9. Device is logged in and the status page opens.
Alternative flows	A1. Payment declined or cancelled: system shows the failure, grants nothing and offers Retry. B1. Payment confirmed but router unreachable: tenant sees Payment received, activating shortly, and the system retries. C1. Service down: Buy a plan is disabled with a notice. D1. Confirmed amount does not match the plan: no access is granted and the order is flagged.
Postconditions	An active subscription exists for the device MAC address until the expiry time.

9.2 UC-02 Tenant checks status

Actor	Tenant
Preconditions	Device is connected.
Main flow	Tenant opens the status page. System identifies the device and shows plan, payment status, start, expiry and remaining time.
Alternative flows	A1. No active subscription: system offers Buy a plan. B1. MAC mismatch: system advises contacting the owner.
Postconditions	None.

9.3 UC-03 Owner approves a tenant's laptop

Actor	Owner
Preconditions	Tenant has paid for a plan and asks for laptop access. Laptop is connected to the Wi-Fi but not logged in.
Main flow	1. Owner logs in to the admin portal. 2. Opens Devices. 3. Selects the tenant's laptop from the list of unauthorised devices. 4. Chooses the tenant name and plan. 5. Confirms. 6. System creates a MAC-bound hotspot user and records the action in the audit log. 7. Laptop gains access.
Alternative flows	A1. Device not in the list: owner asks the tenant to reopen the browser so the router registers the host, then refreshes. B1. Router API error: system shows the error and logs it.
Postconditions	Laptop has an active subscription tied to its MAC address.

9.4 UC-04 Owner re-binds a subscription

Actor	Owner
Preconditions	A tenant's device MAC changed and the old subscription is still active.
Main flow	1. Owner opens Subscriptions and finds the tenant. 2. Selects the new device from the connected devices list. 3. Confirms re-bind. 4. System removes authorisation from the old MAC, moves the hotspot user to the new MAC and keeps the original expiry time. 5. Action is logged.
Alternative flows	A1. Subscription already expired: system advises creating a new one.
Postconditions	Only the new device can use the subscription.

9.5 UC-05 Owner issues a cash voucher

Actor	Owner, Tenant
Preconditions	Tenant has paid the owner in cash.
Main flow	1. Owner generates a voucher for one of the two plans. 2. Owner gives the code to the tenant. 3. Tenant enters the code on the login page. 4. System checks that the voucher is unused, binds it to the device MAC and creates the hotspot user. 5. Voucher is marked used and the action is logged.
Alternative flows	A1. Code already used or expired: system rejects it. B1. Too many failed attempts: system temporarily blocks further tries.
Postconditions	Voucher marked redeemed. Subscription active.

9.6 UC-06 Starlink outage

Actor	System
Preconditions	Internet connection or Fapshi becomes unreachable.
Main flow	1. Health check fails. 2. Portal shows service temporarily down and disables Buy a plan. 3. Existing subscriptions remain valid on the router. 4. When the connection returns, the system reconciles pending orders with Fapshi and provisions any that succeeded.
Alternative flows	A1. Owner issues cash vouchers or approves devices manually during the outage.
Postconditions	No paid tenant is left without access and no unpaid tenant is given access.

9.7 UC-07 Subscription expiry

Actor	System
Preconditions	An active subscription reaches its expiry time.
Main flow	1. Router removes or disables the hotspot user. 2. Internet access is removed within 2 minutes. 3. Scheduled job marks the subscription Expired in the database. 4. Device is redirected to the portal. 5. Tenant can buy another plan.
Postconditions	Subscription is Expired and the device is blocked until a new purchase.

9.8 UC-08 Payment reconciliation

Actor	System (scheduled job)
Preconditions	One or more orders are Pending or Paid without a subscription.
Main flow	1. System queries Fapshi for each such order. 2. System receives the current transaction status. 3. If the payment succeeded, system verifies the amount. 4. System creates the subscription only if the order has not already been applied. 5. System provisions the MikroTik. 6. System records the reconciliation event.
Postconditions	Each successful payment is applied exactly once.

10. Technology Stack and Project Structure

Layer	Choice	Reason
Frontend	HTML, CSS, JavaScript (no framework)	About four pages. Fast on low-end phones. Easy to serve through the walled garden.
Backend	Raw PHP 8.x	Developer's existing skill. Small surface area, no framework overhead.
Web server	Apache or Nginx	Common, well documented, runs on Linux and Windows.
Database	SQLite initially; MySQL/MariaDB if required	SQLite needs no separate server. MySQL is an easy upgrade if needed.
Router	MikroTik RouterOS 7 Hotspot	Built-in captive portal, user profiles, expiry, speed limits and API.
Payments	Fapshi	Supports MTN Mobile Money and Orange Money in Cameroon.
Hosting	Local server on the LAN	No recurring cost. Direct router access. Works when the internet is down.
Tunnel (optional)	Free tunnel service	Gives Fapshi a public HTTPS address for webhooks without opening ports.

The first release prioritises simplicity and low operating cost.

Suggested folder structure

```
portal/
├── public/
│   ├── index.html
│   ├── checkout.html
│   ├── status.html
│   ├── help.html
│   ├── assets/
│   │   ├── style.css
│   │   └── app.js
│   └── api/
│       ├── create_order.php
│       ├── payment_status.php
│       ├── fapshi_webhook.php
│       └── subscription_status.php
├── admin/
│   ├── login.php
│   ├── dashboard.php
│   ├── devices.php
│   ├── subscriptions.php
│   ├── payments.php
│   ├── vouchers.php
│   └── audit.php
├── src/
│   ├── config.php
│   ├── db.php
│   ├── auth.php
│   ├── fapshi.php
│   ├── mikrotik.php
│   ├── subscriptions.php
│   ├── payments.php
│   └── audit.php
├── cron/
│   ├── reconcile.php
│   └── expire.php
├── storage/
│   └── database.sqlite
└── logs/
```

Listing 10.1. Proposed project layout. The src, storage and logs folders sit outside the public web root where possible.

11. Risk Analysis

ID	Risk	Impact	Likelihood	Mitigation
R-01	Tenants share their connection by tethering.	Revenue loss, congestion	High	TTL filter, per-user speed and connection limits, testing, honest documentation of limits.
R-02	Phone MAC address changes and the tenant loses access.	Support effort	Medium	Clear message on the portal, admin re-bind, help page instructions.
R-03	Power cuts stop the router and portal server.	Service interruption	High	UPS or battery backup, automatic start, router-side enforcement.
R-04	Starlink outage during a payment.	Tenant paid without access	Medium	Reconciliation job, outage notice, manual approval by the owner.
R-05	Fapshi API change or downtime.	Payments fail	Low to medium	Isolated integration module, monitoring, pending status and reconciliation, vouchers as fallback.
R-06	Forged or repeated webhook calls.	Free access, duplicate credit	Medium	Verify every transaction with Fapshi, unique transaction identifiers, idempotent processing.
R-07	Router API credentials leaked.	Full control of access	Low	Dedicated low-privilege account, IP restriction, secrets outside web root.
R-08	MAC cloning to use another tenant's subscription.	Account abuse	Low	One session per user limits impact. Audit log helps the owner investigate.
R-09	Captive-portal pop-up does not appear on some phones.	Tenant confusion	Medium	Help page with manual address, test across common phones before launch.
R-10	Private DNS or VPN on tenant phone blocks the redirect.	Cannot reach the portal	Medium	Help guidance to switch these off while signing in.
R-11	Single server failure.	No new sales	Medium	Regular backups, simple reinstall guide, replaceable hardware, router-side enforcement.
R-12	Browser manipulates plan or amount.	Underpayment	Medium	Server-side price, amount check against Fapshi, rejection of mismatches.

12. Testing and Acceptance Criteria

12.1 Test approach

- Unit and module tests: PHP functions for order creation, status handling, amount checks, duplicate protection and router calls, using a mocked Fapshi and a test router profile.
- Integration tests: full payment flow in Fapshi's sandbox or with small real amounts for both plans, and router provisioning on a real MikroTik.

- Device tests: at least two Android devices from different manufacturers, an iPhone, a Windows laptop and, where available, a macOS laptop, including devices with private or randomised Wi-Fi MAC addresses.
- Failure tests: Starlink disconnected, MikroTik unavailable, portal server switched off, Fapshi unavailable, payment declined, payment pending, repeated payment notification, server restart, router restart, power interruption.
- Security tests: SQL injection, XSS, unauthorised admin access, parameter manipulation, payment replay, forged payment notifications, access before payment, second-device access, router API exposure, credential exposure, and tethering attempts.
- Functional checks: unpaid device is blocked, wrong amount is rejected, failed payment gives no access, subscriptions expire correctly, voucher works once, revocation and extension work.

12.2 Acceptance criteria

ID	Criterion	Related requirements
AC-01	An unpaid device cannot reach any external website or app, and can open only the permitted portal resources.	FR-CP-01, FR-CP-02, FR-CP-05
AC-02	After confirmed payment, the subscription is provisioned and internet access is available within 10 seconds, subject to external service and network response time.	FR-PY, FR-AP-03, FR-AP-04, FR-AP-06
AC-03	A successful 2,000 XAF payment creates a 10-day subscription and no other plan.	FR-PL-02, FR-PL-05
AC-04	A successful 5,000 XAF payment creates a 30-day subscription and no other plan.	FR-PL-03, FR-PL-05
AC-05	A second device cannot use a subscription bought for the first device.	FR-DB-01 to FR-DB-03
AC-06	Standard Wi-Fi and USB tethering tests are blocked or restricted according to the defined anti-tethering test cases.	FR-AT-01, FR-AT-02, FR-AT-08
AC-07	A repeated or forged payment notification creates no second subscription and grants no unauthorised access.	FR-PY-04, FR-PY-06, FR-AP-10, NFR-SC-06
AC-08	An expired subscription loses access within 2 minutes of expiry.	FR-AP-05, NFR-PF-04
AC-09	The owner can re-bind an active subscription to another device and the previous device loses authorisation.	FR-DB-05, FR-AD-06
AC-10	The owner can approve a connected laptop in under one minute.	FR-AD-02, FR-AD-03
AC-11	The owner can revoke an active device or subscription immediately.	FR-AD-04, FR-AD-09
AC-12	With the portal server switched off, existing subscribers stay online until expiry.	FR-OU-05, FR-AP-05
AC-13	With Starlink unplugged, the portal does not report internet as available and disables payments. After reconnection, a payment started earlier is reconciled.	FR-OU-01 to FR-OU-04
AC-14	Fapshi credentials do not appear in browser source, client-side JavaScript, responses or normal application logs.	FR-PY-03, FR-LG-02, NFR-SC-01
AC-15	A tenant cannot access the administration interface.	FR-AD-11, NFR-SC-08
AC-16	A cash voucher works once, creates the appropriate subscription and cannot be reused.	FR-VC-01, FR-VC-02
AC-17	Database records survive a normal portal-server restart.	Section 8.3
AC-18	The database can be restored from a valid backup.	Section 8.3, section 13.2
AC-19	A payment with a wrong amount, or one that failed, was cancelled or expired, grants no access.	FR-PL-06, FR-PL-07, FR-PY-11, FR-PY-12

13. Deployment and Operations

13.1 Deployment steps

1. Install and test the Starlink dish, then enable bypass mode on its router where applicable.

2. Connect Starlink to the MikroTik WAN. Configure the LAN bridge, DHCP, NAT, time synchronisation and firewall basics.
3. Set up the Hotspot with MAC login, one profile per plan, the calendar-expiry method, walled garden and the TTL filter.
4. Install the web server, PHP and database on the local portal server and give it a reserved LAN address.
5. Deploy the portal files, configuration file (Fapshi and router credentials) and database schema.
6. Create the two subscription plans in the database.
7. Create the router API account and restrict it to the portal server.
8. Customise the hotspot login page and point it to the portal.
9. Test the payment flow, device binding, subscription expiry, administrator functions and failure scenarios against the acceptance criteria, using several phones and laptops.
10. Set up automatic backups, scheduled reconciliation and expiry jobs.
11. Train the owner on the admin portal, then open the service to tenants.

13.2 Ongoing operations

Task	Frequency
Check the dashboard for failed, stuck payments and provisioning errors	Daily
Reconcile Fapshi transactions with subscriptions	Daily (automatic) and weekly (manual review)
Back up the database	Daily
Back up the router configuration	After each router change
Keep one backup separate from the portal server	Ongoing
Review the audit log	Weekly
Review service and network, and measure Starlink throughput	Monthly
Update PHP, web server and RouterOS	Planned maintenance, after reading release notes
Test restoring a backup	Every three months

14. Future Enhancements

- Tenant self-service extra-device add-on with its own price.
- SMS or WhatsApp reminders before a subscription expires.
- Additional or changed plans, data-volume plans and a daily fair-use cap.
- Plan-management screen in the admin portal.
- Revenue and usage charts for the owner, with monthly reports.
- Support for several routers or buildings from one admin portal.
- Referral discounts and promotional codes.

- Automatic renewal prompts at expiry and customer accounts.
- Native mobile application.

15. Appendix A: Traceability Summary

The table links each business objective to the requirement groups that deliver it, so reviewers can check that every objective is covered.

Business objective	Requirement groups
Tenants can pay by mobile money without owner involvement	FR-CP, FR-PL, FR-PY, FR-AP, NFR-PF-01, NFR-PF-02
Two simple, fixed subscription options	FR-PL-01 to FR-PL-09
Access lasts for exactly the period paid for	FR-PL-05, FR-AP-05, NR-06, NFR-PF-04
One payment covers one device only	FR-DB, FR-AT, NR-04, NR-05, NR-09
Owner can serve special cases such as laptops and cash payments	FR-AD, FR-VC, UC-03 to UC-05
Payments are trustworthy and cannot be faked or duplicated	FR-PY-04, FR-PY-06, FR-PY-11, FR-AP-10, NFR-SC-06, AC-07, AC-19
Service degrades safely during outages and power cuts	FR-OU, section 8.3, R-03, R-04, AC-12, AC-13
Tenant and payment data is handled responsibly	Section 6.4, NFR-SC, FR-LG, FR-ST-02
System is simple to run, low cost and easy to maintain	Section 2.6, section 8.5, sections 10 and 13
System runs without cloud hosting	Section 2.6, section 10

16. Appendix B: Items to Confirm Before Development

- Plan speed profiles for the 10-Day and 30-Day Plans (OD-1).
- Fapshi payment method (hosted checkout or direct charge) and exact API details: endpoints, authentication, payment initiation, status method, webhook behaviour, supported networks and status values (OD-2, OD-3).
- Router model and RouterOS version (OD-4).
- Starlink dish and router generation and Ethernet arrangement (OD-5).
- Portal server hardware, selected and tested (OD-6).
- Whether English only or English and French interfaces are required at launch (OD-7).
- Data retention period and wording of the privacy and terms notices (OD-8).

17. Approval

This SRS becomes the baseline for development once the open decisions in section 2.7 have been confirmed.

Role	Name	Signature	Date
Project Owner	Mbuh Favour	_____	_____
Developer	_____	_____	_____
Network Technician	_____	_____	_____
Tester	_____	_____	_____

— END OF SOFTWARE REQUIREMENTS SPECIFICATION —

Starlink Wi-Fi Subscription Portal — Version 1.2